1. Clinical Question & Core Hypothesis

Clinical Question

The provided path shows PROS1 is expressed in tumors and associated with Ki-67, raising the question of why immune organization seen on H&E (e.g., TLS/TIL) does not always translate into better clinical outcomes — could tumor-expressed PROS1 drive proliferation/immune modulation that negates lymphoid aggregate benefit?

Working Hypothesis

Within TCGA-BRCA, TLS-rich breast tumors that also have high tumor-derived PROS1 expression and high MKI67 (Ki-67) exhibit a net pro-proliferative and immune-modulatory phenotype driven by PROS1-TAM receptor signaling, which compromises TLS-associated anti-tumor activity and leads to worse overall survival (OS) compared with TLS-rich tumors with low PROS1 expression. This implies a PROS1–MKI67 axis that links tumor PROS1 to proliferation and macrophage-mediated immune modulation, such that: (1) among TLS-high tumors, PROS1 is positively correlated with MKI67; (2) TLS-high/PROS1-high tumors have poorer OS than TLS-high/PROS1-low tumors after adjusting for covariates (age, stage, subtype, tumor purity, etc.); (3) a TAM-receptor signaling signature mediates this association; and (4) in vitro assays show that PROS1 overexpression or PROS1-containing conditioned media increases proliferation and skews macrophages toward a pro-tumor (M2-like) phenotype in a MerTK-dependent manner. This framework yields testable predictions and supports a mechanistic link between PROS1, tumor proliferation, TLS architecture, and clinical outcome in breast cancer.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PROS1, MKI67
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PROS1: PROS1 drive proliferation/immune modulation that negates lymphoid aggregate benefit? (2) Tier 1 path used: the single, strongest path provided (Tumors → PROS1 → Ki-67) — I built on this because it directly links a tumor-derived protein to proliferative state, which plausibly modifies immune-related imaging signals.

MKI67: MKI67 axis as a modifier of an imaging-derived immune feature.

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions